

Part 1:

Assuming the following initial data in `t_employee`:

c_eid	c_name	c_department_id	c_salary	c_hire_date
1	Alice	1	50000	2020-01-15
2	Bob	2	60000	2019-03-10
3	Carol	3	45000	2021-07-23
4	David	2	70000	2018-05-22
5	Eve	4	55000	2017-08-19
6	Frank	5	48000	2019-11-30
7	Grace	5	50000	2021-01-01
8	Hannah	5	52000	2020-02-17

ALTER TABLE Questions:

1. Add a new column:

How would you add a new column `c_email` of type `VARCHAR(100)` to the `t_employee` table?

2. Modify a column type:

How would you modify the `c_name` column to increase the length to `VARCHAR(100)`?

3. Rename a column:

How would you rename the `c_department_id` column to `c_dept_id`?

4. Drop a column:

How would you remove the `c_hire_date` column from the `t_employee` table?

5. Set a default value:

How would you set a default value of 0 for the `c_salary` column in the `t_employee` table?

6. Add a unique constraint:

How would you add a unique constraint on the `c_email` column to ensure no two employees have the same email?

7. Drop a unique constraint:

How would you drop the unique constraint on the c_email column if it was added?

8. Add a foreign key constraint:

How would you add a foreign key constraint on the c_department_id column, referencing a t_department table?

9. Drop a foreign key constraint:

How would you drop the foreign key constraint on the c_department_id column?

10. Add a NOT NULL constraint:

How would you ensure that the c_name column does not accept NULL values?

UPDATE Questions:

11. Update a specific row:

How would you update the salary of 'Alice' to 55000?

12. Update multiple rows:

How would you increase the salary of all employees in department 5 by 5%?

13. Conditional update:

How would you set the c_hire_date to '2021-01-01' for all employees whose salary is less than 50000?

14. Update with a JOIN:

How would you update the c_department_id for all employees based on a t_department table where department names match?

15. Set NULL value:

How would you set the c_hire_date to NULL for all employees hired before '2018-01-01'?

16. Update with a subquery:

How would you increase the salary of employees by 10% if their salary is less than the average salary of all employees?

RENAME Questions:

17. Rename a table:

How would you rename the t_employee table to t_staff?

18. Rename a constraint:

How would you rename a primary key constraint on the c_eid column to pk_employee_id?

19. Rename a sequence:

How would you rename the sequence for the SERIAL column c_eid if it was named t_employee_c_eid_seq?

DELETE Questions:

20. Delete a specific row:

How would you delete the row of the employee named 'David'?

21. Delete with a condition:

How would you delete all employees from department 4?

22. Delete all rows:

How would you delete all rows from the t_employee table but keep the table structure intact?

23. Delete using a JOIN:

How would you delete all employees who belong to a department that has been removed from the t_department table?

24. Delete with a subquery:

How would you delete employees whose salary is below the average salary of all employees?

DROP TABLE Question:

25. Drop the table:

How would you drop the t_employee table, ensuring that it no longer exists in the database?

Part 2:

1. Basic Queries:

1. Select all employees
2. Find employees who earn more than 50,000
3. List all employees from a specific department (e.g., department 2)
4. Find employees hired after January 1st, 2020
5. Find employees with the same salary
6. Find employees whose name starts with 'A'
7. List employees whose salary is between 45,000 and 60,000
8. Find employees hired between two dates
9. Find employees who earn the same as or more than the average salary
10. Find employees hired more than 5 years ago
11. Find all employees who were hired on a Monday
12. List employees and their hire year

2. Aggregations:

1. Get the total number of employees
2. Calculate the average salary of all employees
3. Get the highest salary in the company
4. Get the minimum salary in the company
5. Find the total salary cost for a specific department (e.g., department 5)
6. Get the total salary of employees hired after 2020
7. Find the second-highest salary

3. Grouping and Sorting:

1. Group employees by department and find the total salary per department
2. Count the number of employees per department
3. Get the department with the highest total salary
4. List employees ordered by hire date (oldest first)
5. List employees ordered by salary in descending order

6. Find employees with a salary higher than the average salary in their department

4. Conditional Operations:

1. Get employees who were hired between two dates and have a salary above a certain threshold
2. Find employees whose salary is below the average salary and were hired after a specific date